Yunchun CUI

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Research Interests

Robotic Manipulation, Dexterous Hands, Vision-Tactile Perception

Education

B.Eng. in Energy and Power Engineering Sept. 2023 – Jun. 2027 (expected)
Jilin University, Changchun, China
College of Automotive Engineering
GPA: 3.62/4.0 (top 15%)
Relevant coursework: Fundamentals of Mechanical Design A (89), The Control Engineering Basic Theory (92.3), Probability and Statistic B (90.5), Production Practice of Mechanical Engineering (91)

Research Experience

Underwater Dexterous Manipulation with Adaptive Impedance Control Jul. 2026 – Present
Student Researcher
Supervisor: Prof. Chunbao Liu, Jilin University

- Designed a neural hydrodynamic disturbance observer (NHDO) for online 6-DoF water-jet wrench estimation from proprioceptive and jet-geometry features, with bounded-error analysis of the estimation.
- Developed a disturbance-compensated adaptive impedance controller with saturated gain adaptation and proved uniform ultimate boundedness (UUB) of the closed-loop system for a torque-less, position-controlled robotic arm.
- Implemented the control stack on real hardware (Realman RML63III-6FB arm, waterproof WA100 hand, wrist six-axis F/T), with the admittance control loop running at 20 Hz through the arm's position interface; achieved an RMS hydrodynamic-wrench estimation error of **0.07 N** (**10%** of the mean disturbance).
- Performed **grasp and jet-force calibration and real-world experiments**; staged thumb–finger calibration achieved **36/36 (100%)** successful plate grasps, while water-jet disturbances were characterized at two flow settings (**0.37±0.07 N** at 7.02 L/min and **0.71±0.09 N** at 8.75 L/min).
- Reduced jet-induced end-effector deviation by **90%**, from 17.8 mm without compensation to 1.8 mm with compensation, under the 0.71 N jet condition.

Hierarchical Neural Network Control with Adaptive Barrier Activation for Constrained Robotic Manipulators May. 2026 – Present
Student Researcher
Supervisor: Prof. Chunbao Liu, Jilin University

- Co-developed the MATLAB simulation framework for a 6-DOF RML63II-6F manipulator, implementing prescribed performance control, BLF-based constraint handling, and dual-layer RBF neural networks (25/20 nodes).
- Performed controller tuning and comparative validation under 6 kg payload, ±50 Nm torque saturation, sensor noise, actuator dynamics, and 200 Hz control, benchmarking against five baseline controllers.
- Conducted 50-trial Monte Carlo validation, achieving 1.67 mm average tracking error, 0.2% constraint violation rate, and 18.3% barrier-layer activation, with 21.9% lower tracking error and 28.5% lower energy consumption than Dual-NN-Binary.

Vision-Based Traversability Mapping for Lunar Rover Navigation Apr. 2026 – Aug. 2026
Student Researcher
Supervisor: Prof. Wenchao Xu, Jilin University

- Built a lunar terrain dataset of 726 images with 3,586 annotated instances across 9 geomorphological classes and trained a YOLOv8 crater detector; the frame-level hazard decision used by the gait controller achieved **92.1% recall** at **68.6% precision** on the held-out test set, compared with **96.3% recall** on the validation set.
- Implemented real-time inference at **127 FPS** (7.9 ms/frame, 960×960 input) with 3-frame detection skipping and a **10 Hz gait-control interface** for adaptive locomotion.
- Developed a patented complex-terrain perception pipeline integrating adaptive image enhancement, multi-feature extraction, and uncertainty-aware traversability cost mapping with ground-frame projection.

Innovative Design of Vehicle Bionic Suspension System Mar. 2026 – Jul. 2026
Student Researcher
Supervisor: Prof. Wenchao Xu, Jilin University

- Completed the integrated structural design of braking, steering, and suspension systems using AutoCAD.
- Filed an invention patent as the second inventor (granted).

Generative AI-Driven Inverse Design of CFRP/Steel Composite Crash Beams Jan. 2025 – Aug. 2026
Student Researcher
Supervisor: Prof. Wenchao Xu, Jilin University

- Developed a **CVAE-based generative design framework** integrating machine learning with FE-based structural analysis for inverse design of CFRP/steel composite crash beams.
- Reduced design-performance MAE from **0.7910 to 0.4864** and increased generation diversity from **0.0831 to 0.2122**; completed **structural prototype fabrication**.

Research Intern May. 2026 – Present
Lingyan Technology

- Investigated **sim-to-real simulation methods for deformable silicone skin** and facial motion modelling.
- Developed a **BlendShape-based facial simulation model** supporting configurable motion-control points in both number and spatial location.
- Built a **full upper-body simulation environment** with a **ROS 2** communication pipeline for real-time transmission of point coordinates and motion-control commands.

Competitions

China Undergraduate Engineering Practice and Innovation Competition Sep. 2026

- Designed the cam-actuated steering mechanism and payload stabilization system for a biomass-powered autonomous vehicle, including SolidWorks-based modeling of the cam-linkage structure and development of a spring-loaded ball holder for stable operation on ramps and uneven surfaces.

National University Students Competition on Energy Conservation and Emission Reduction Apr. 2026
National Second Prize

- Developed the CATIA-based 3D model of a biomimetic battery thermal management system inspired by tree-root and butterfly-wing structures, focusing on the design of hierarchical branching cooling channels and structural integration.

Awards & Honors

- Second Prize, National University Students Competition on Energy Conservation and Emission Reduction, 2026
- Provincial Outstanding Project, Undergraduate Innovation and Entrepreneurship Training Program, 2025
- 2nd Class Scholarship, Jilin University, 2025
- 1st Class National Financial Aid, 2025 and 2024

Technical Skills

Programming: Python, C++, MATLAB
Robotics & Simulation: NVIDIA Isaac Sim, ROS/ROS 2, MuJoCo
Machine Learning & Vision: PyTorch, OpenCV, NumPy, Pandas
Mechanical Design: CATIA, SolidWorks
Development Tools: Git, Linux, LaTeX, Jupyter
Languages: English (IELTS 6.5), Chinese (Native)

Publications

Journal Papers

- Y. Cui, et al. "Performance Analysis and Optimization Design of Integrated Battery Box Structure for Electric Cargo Vehicles." *Engineering Research Express*, accepted, 2026.

Manuscripts Under Review / In Preparation

- **Y. Cui** (Co-first author), et al. "Neural Hydrodynamic Disturbance Observation and Adaptive Impedance Control for Waterproof Dexterous Hand–Arm Manipulation under Water-Jet Disturbances." Manuscript in preparation, 2026.
- **Y. Cui** (Second author), et al. "Hierarchical Neural Network Control with Adaptive Barrier Activation for Constrained Robotic Manipulators." Under review at *Expert Systems with Applications*, 2026.
- **Y. Cui** (Second author), et al. "Generative AI-Driven Integrated Design of Automotive Carbon/Steel Composite Crash Beam." Under review at *Advanced Engineering Informatics*, 2026.

Patents

- 1st Inventor, "A Method and System for Image Enhancement and Traversability Cost Map Generation in Complex Environment," granted, Aug. 2026.
- 2nd Inventor, "A Vehicle Corner Module," granted, Jul. 2026.
- 5th Inventor, "Intelligent Generative Design Method for Anti-Collision Beams," granted, Oct. 2025.

References

Available upon request.